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on

"Telco Customer Churn: Exploratory and Confirmatory Data Analysis"

Team Number: Team_02

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1. Introduction

Customer churn, the loss of customers who stop using a service, is a central concern for subscription businesses such as telecom operators, so knowing which customers leave, and when, is of clear practical interest. This mini project analyses the public "Telco Customer Churn" data set [1], [2] in two parts: **Exploratory Data Analysis (EDA)** describes churn patterns in the data, and **Confirmatory Data Analysis (CDA)** formally tests whether churn depends on the type of customer contract.

2. Objectives and Problem Statement

Problem. Do customers on different contract types churn at different rates, and which other customer characteristics go together with leaving? **Objectives:** (i) assess data quality and prepare the data; (ii) describe churn across tenure, charges, contract and other attributes (EDA); (iii) test whether churn is independent of contract type and measure the strength of any association (CDA); (iv) interpret the findings and state clearly what observational data can and cannot show.

3. Dataset and Methodology

3.1 Data Collection: Source and Description

We used the public data set "Telco Customer Churn" [1] (file WA_Fn-UseC_-Telco-Customer-Churn.csv), whose description credits the IBM sample data sets. IBM's documentation describes a fictional telecom company and states that the churn column records whether a customer left within the last month [2]. The file was used exactly as supplied, so findings describe this data set and should not be generalised to a real operator. Each of the 7,043 rows is one customer and each of the 21 columns an attribute (Table 1). The target variable *Churn* is "Yes" for 1,869 customers (26.54%) and "No" for 5,174 (73.46%).

Table 1. Variables of the data set.

Group	Variables	Values / summary
Identifier	customerID	Unique key (not analysed)
Demographics	gender, SeniorCitizen, Partner, Dependents	Binary attributes
Account	tenure, Contract, PaperlessBilling, PaymentMethod	Tenure 0–72 months (mean 32.37, median 29); 3 contract types; 4 payment methods
Charges	MonthlyCharges, TotalCharges	Monthly: 18.25–118.75 (mean 64.76); total: 0–8,684.80 (mean 2,279.73); currency as in file
Services	PhoneService, MultipleLines, InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies	Yes / No, plus "No phone service" or "No internet service"; internet: DSL / Fibre optic / No
Target	Churn	Yes = 1,869; No = 5,174

3.2 Data Preprocessing

Each data-quality check, its result and the action taken are listed in Table 2.

Table 2. Data-quality checks and preprocessing actions.

Check	Result	Action
Size and types	7,043 rows × 21 columns; *TotalCharges* was read as text, other types as expected	Converted *TotalCharges* to numeric
Missing values	0 NaN; 11 blank *TotalCharges* cells, all with tenure = 0 (new customers not yet billed; all Churn = No)	Set to 0 (nothing billed yet); no rows dropped
Duplicates	0 duplicate rows and 0 duplicate IDs; 22 rows equal to another row except the ID	Kept: distinct customers may share attributes
Invalid values	No negatives; tenure within 0–72; category labels as expected; "No phone / internet service" consistent	None needed
Variable roles	*customerID* is a unique key; *SeniorCitizen* is coded 0/1	ID excluded; *SeniorCitizen* treated as categorical
Outliers	*TotalCharges* is right-skewed (Fig. 2) but values are valid	No values removed

After repair the analysis table has 7,043 complete rows and 20 usable columns.

3.3 Tools and Technologies Used

Analysis was done in Python in a Jupyter notebook (Telco_Churn_Analysis.ipynb): pandas [3] for data handling, SciPy [4] for the statistical tests and Matplotlib [5] for the figures. The notebook, data and outputs are available in the project repository (link at the end of this report); the notebook also contains a supplementary predictive-modelling section that goes beyond the scope of this report.

4. Literature Review / Background Study

Churn in telecommunications has been studied mainly through predictive modelling. Verbeke et al. [6] benchmarked classification techniques on eleven telecom data sets from operators worldwide and found that a small number of variables suffices to predict churn with high accuracy and that oversampling generally does not improve performance significantly; they also argued that models should be judged by the profit they generate rather than by statistical measures alone. Because churn is usually a rare event, Burez and Van den Poel [7] examined how class imbalance can be handled in churn prediction. De Caigny et al. [8] noted that decision trees handle linear relations poorly whereas logistic regression handles interaction effects poorly, and proposed a hybrid "logit leaf model" benchmarked against decision trees, logistic regression and random forests. Ahmad et al. [9] built a churn model for a telecom operator on a big-data platform; XGBoost performed best, and adding social-network features raised the AUC from 84% to 93.3%. Ascarza et al. [10] ran a large field experiment in which offering customers plan recommendations increased churn (10% against 6% in the control group), showing that proactive retention campaigns can raise rather than lower churn.

Relation to this project. These studies concentrate on prediction, often with proprietary or very large data, whereas this project describes and formally tests an association on a small public data set. Reference [6] indicates that a few attributes carry much of the predictive signal, which is consistent with contract type and tenure standing out in our EDA. Reference [7] is relevant because our data show a moderate class imbalance (26.54% churn), which we leave unaltered in the EDA and the chi-square analysis. Reference [8] motivates

logistic regression as the multivariable follow-up proposed in Section 7.3, and [10] reinforces our caution that an association between contract type and churn does not show that changing contracts would reduce churn.

5. Implementation: Exploratory Data Analysis (EDA)

Churn rate. Figure 1a shows 1,869 churners and 5,174 retained customers (26.54% vs 73.46%; ratio 2.77 : 1), a moderate class imbalance.

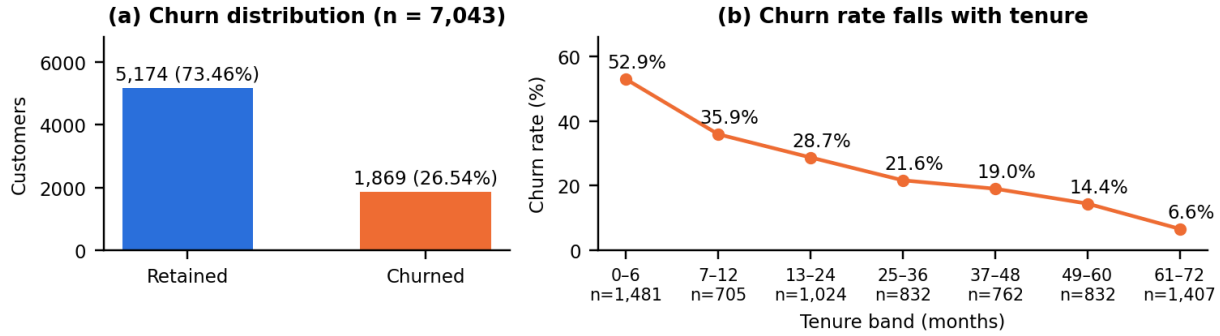


Figure 1. (a) Distribution of *Churn* (n = 7,043). (b) Churn rate by tenure band (months).

Numerical variables. Churners have a much shorter median tenure (10 vs 38 months), a higher median monthly charge (79.65 vs 64.43) and a lower median total charge (703.55 vs 1,679.53), the last mainly because they have been customers for less time (Fig. 2). Churn falls steadily with tenure, from 52.94% in the first six months to 6.61% after five years (Fig. 1b). Spearman correlations: tenure–*TotalCharges* $\rho = 0.89$, churn–tenure $\rho = -0.37$, churn–*TotalCharges* $\rho = -0.23$, churn–*MonthlyCharges* $\rho = 0.18$.

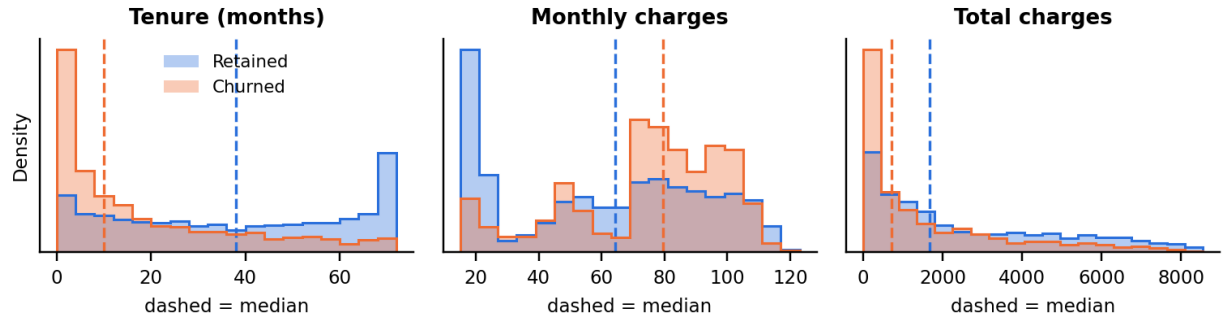


Figure 2. Distributions of tenure, monthly charges and total charges by churn status (dashed lines = medians).

Categorical variables. Churn is highest for month-to-month contracts (42.71%, vs 11.27% for one-year and 2.83% for two-year), fibre-optic internet (41.89%; DSL 18.96%; no internet 7.40%), electronic-check payment (45.29%; other methods 15.24–19.11%) and paperless billing (33.57% vs 16.33%) (Fig. 3). Senior citizens churn at 41.68% (others 23.61%), customers without a partner at 32.96% (with: 19.66%) and without dependents at 31.28% (with: 15.45%). Gender makes almost no difference (female 26.92%, male 26.16%).

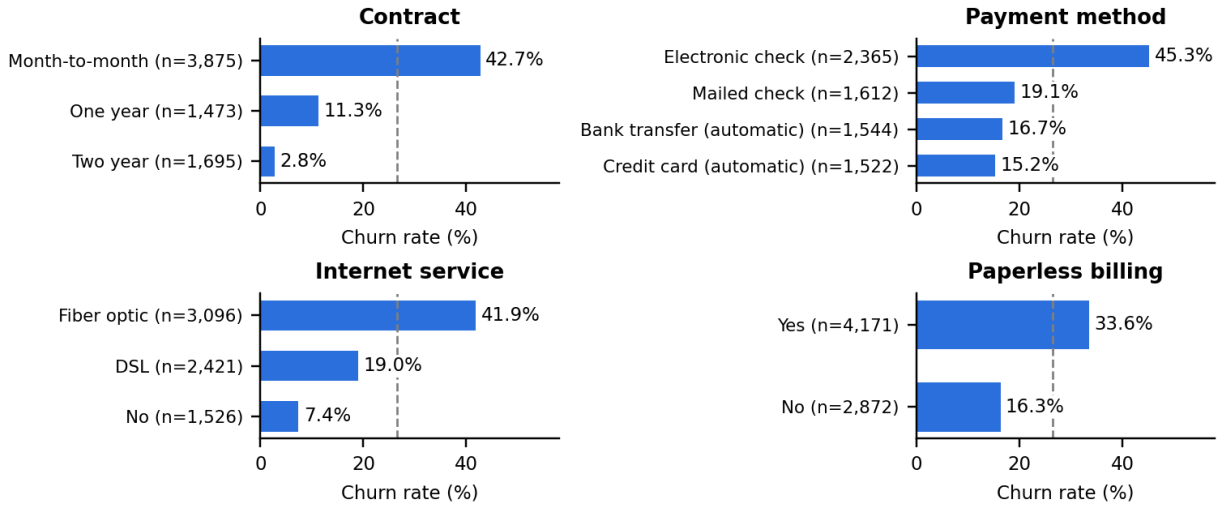


Figure 3. Churn rate (%) by contract, payment method, internet service and paperless billing (dashed line = overall 26.54%).

EDA summary. Short tenure and a month-to-month contract go together with much higher churn. Because contract type shows the widest spread of churn rates among the account attributes, it is tested formally in Section 6.

6. Confirmatory Data Analysis (CDA)

6.1 Test and Hypotheses

Contract type (3 categories) and churn (2 categories) are both categorical, so a Pearson chi-square test of independence [11] on the 3 × 2 contingency table is appropriate. Significance level $\alpha = 0.05$.

H_0: contract type and churn are independent (the churn probability is the same for all three contract types).

H_1: contract type and churn are not independent (the churn probability differs for at least one contract type).

6.2 Observed Contingency Table

Table 3. Contract type × churn: observed counts [expected counts under H_0] and churn rate with 95% Wilson interval [12].

Contract	Retained (No)	Churned (Yes)	Total	Churn rate (95% CI)
Month-to-month	2,220 [2,846.7]	1,655 [1,028.3]	3,875	42.71% (41.16–44.27)
One year	1,307 [1,082.1]	166 [390.9]	1,473	11.27% (9.75–12.99)
Two year	1,647 [1,245.2]	48 [449.8]	1,695	2.83% (2.14–3.73)
Total	**5,174**	**1,869**	**7,043**	**26.54%**

6.3 Assumptions

(i) Both variables are categorical with mutually exclusive categories. (ii) Every customer appears once (no duplicate IDs), so observations are treated as independent; the sampling design of the (fictional) data cannot be checked. (iii) All expected counts are at least 5; the smallest is 390.9 (one-year, churned). The assumptions are therefore satisfied; the table is 3 × 2, so no continuity correction applies and no alternative test (Fisher's exact test, category merging) is needed.

6.4 Results

Table 4. Chi-square test of independence between contract type and churn.

Quantity	Value
Test statistic	$\chi^2 = 1{,}184.60$
Degrees of freedom	$(3 - 1) \times (2 - 1) = 2$ (critical value $\chi^2_{0.05,2} = 5.99$)
p-value	5.86×10^{-258} (< 0.001)
Effect size (Cramér's V [13])	$V = \sqrt{[\chi^2 / (N \cdot \min(r-1, c-1))]} = \sqrt{(1{,}184.60 / 7{,}043)} = 0.410$
Decision	Reject H_0 at $\alpha = 0.05$

Interpretation. Contract type and churn are strongly associated: $\chi^2(2, N = 7{,}043) = 1{,}184.60$, $p < 0.001$, $V = 0.41$. With $\min(r-1, c-1) = 1$, Cohen's benchmarks [14] are 0.10 (small), 0.30 (medium) and 0.50 (large), so the effect is medium to large. Because N is large, the effect size and the churn rates matter more than the tiny p-value. The likelihood-ratio test agrees ($G^2 = 1{,}386.8$, df = 2, $p < 0.001$).

Where the difference lies. Adjusted standardised residuals (Fig. 4b) show more churners than expected under independence for month-to-month contracts (+34.0) and fewer for one-year (−14.9) and two-year contracts (−25.4); all exceed ±1.96. Pairwise 2 × 2 tests (Yates-corrected, Bonferroni-adjusted for 3 comparisons) are all significant: month-to-month vs one-year $\chi^2(1) = 468.41$, $p_{adj} = 2.1 \times 10^{-103}$; month-to-month vs two-year $\chi^2(1)$

= 881.55, $p_{adj} = 3.0 \times 10^{-193}$; one-year vs two-year $\chi^2(1) = 87.74$, $p_{adj} = 2.2 \times 10^{-20}$. The odds of churning are 25.6 times higher on a month-to-month than on a two-year contract (95% CI 19.1–34.3).

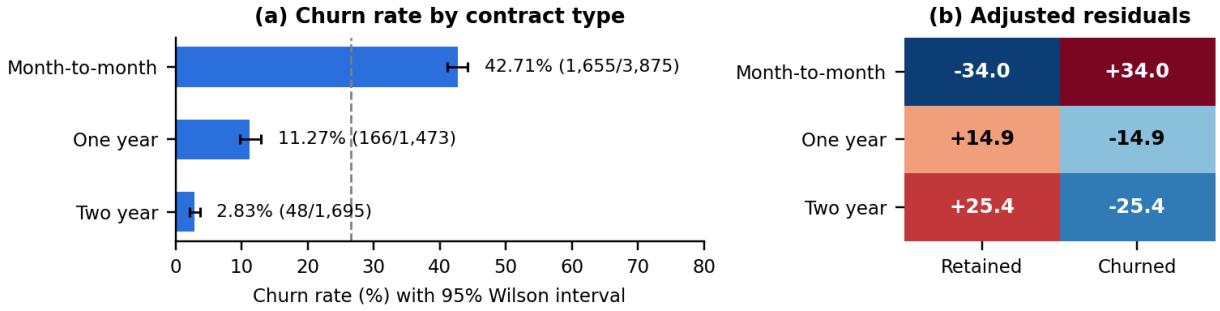


Figure 4. (a) Churn rate by contract type with 95% Wilson intervals. (b) Adjusted standardised residuals of the contingency table (positive = more than expected).

Caution. This is an association in observational data, not proof that the contract itself causes churn. Month-to-month customers also have shorter tenure (median 12 vs 44 and 64 months) and more often use fibre-optic internet (54.9% vs 36.6% and 25.3%), both of which are linked to churn; the test does not adjust for them. For context, among all 16 categorical features screened against churn, *Contract* had the largest Cramér's V (next: *OnlineSecurity*, 0.347).

7. Results, Discussion, Conclusion and Future Work

7.1 Principal Results

- **Data.** 7,043 customers, 21 attributes, 26.54% churn. The data were clean except for 11 blank *TotalCharges* values (new customers), which were set to 0.
- **EDA.** Churn is concentrated early in the customer relationship: 52.94% churn in the first six months of tenure vs 6.61% after five years (median tenure of churners 10 months vs 38). Fibre-optic internet, electronic-check payment, paperless billing and senior-citizen status are also associated with higher churn; gender is not.
- **CDA.** Churn is not independent of contract type: $\chi^2(2, N = 7{,}043) = 1{,}184.60$, $p < 0.001$, Cramér's $V = 0.41$; churn is 42.71% (month-to-month), 11.27% (one-year) and 2.83% (two-year), and all pairwise differences are significant after Bonferroni adjustment.

7.2 Discussion and Limitations

The results meet the objectives: the EDA locates churn among new and month-to-month customers, and the CDA confirms that the contract–churn association is far beyond chance and of medium-to-large size. Limitations: the data are a fictional IBM sample, observational and cross-sectional, and the chi-square test does not control for tenure or services, so confounding is possible.

7.3 Conclusion and Future Work

Contract type is strongly associated with churn, and churn is concentrated among new and month-to-month customers, who are therefore a sensible focus for retention efforts. Whether offering longer contracts would actually reduce churn cannot be concluded from these data. **Future work:** a stratified or multivariable inferential analysis (for example logistic regression) to adjust for tenure and service type, and validation on real,

time-stamped data. The repository notebook also contains a supplementary predictive logistic regression, which is not used for inference in this report.

7.4 Team Member Contributions

Team member	Contribution
Lavanya Sandeep Borkar	Confirmatory data analysis, Figure 4 and Tables 3–4 (Section 6); results, conclusion and future work (Section 7); references and final formatting.
Tanvi Choudhary	Literature review (Section 4); exploratory data analysis and Figures 1–3 (Section 5).
Gauri Nikhil Chandak	Introduction, objectives, dataset description and data preprocessing (Sections 1–3); data-quality checks (Table 2).

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Project repository (code, data and outputs): https://github.com/alien-2709/DA-MINI-PROJECT